

Binding Allocates States; Catalysis Acts from Those States

A model class that joins biological questions to analysis

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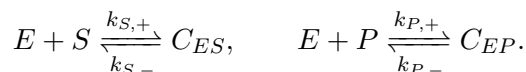
CCBS 2026 Fall — Lecture 8, Sections 1–2

Binding allocates molecular states

Binding does not directly produce the output. It first decides how much of each molecule occupies each possible molecular state.

1. Start from the simplest binding reaction

The enzyme E can bind either substrate S or product P :



Thus the enzyme may be free, substrate-bound (C_{ES}), or product-bound (C_{EP}). Each association–dissociation pair allocates enzyme among these states; it does not yet convert substrate-form material into product-form material.

2. What does “allocates states” mean?

Binding neither creates nor destroys enzyme or substrate. It redistributes each constituent between free and bound states:

- enzyme is free E , substrate-bound C_{ES} , or product-bound C_{EP} ;
- substrate is free S or in C_{ES} ;
- product is free P or in C_{EP} .

Therefore the total inventories are

$$q_E = E + C_{ES} + C_{EP}, \quad q_S = S + C_{ES}, \quad q_P = P + C_{EP}.$$

The individual state concentrations can change, but these totals do not change through binding alone.

3. Binding is not chemical conversion

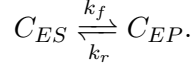
In either binding pair, substrate remains substrate-form and product remains product-form; binding only changes which species carries each constituent. Binding answers *which states are occupied*. It does not yet convert substrate into product or generate the functional output.

Catalysis acts from states

Binding determines how much occupies each state; catalysis determines what those occupied states do.

1. Catalysis occurs from a particular bound state

For the reversible, product-binding enzyme, chemical conversion occurs between the two bound complexes:



The forward channel converts substrate-form material to product-form material; the reverse channel does the opposite. Here $k_f = k_{\text{cat}}^+$ and $k_r = k_{\text{cat}}^-$ are catalytic frequencies, each with units of inverse time.

2. State concentration determines catalytic flux

The forward and reverse fluxes are the concentration of their identified source state multiplied by its catalytic frequency:

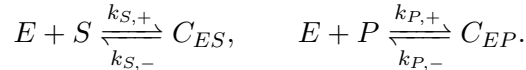
$$v_f = k_f C_{ES}, \quad v_r = k_r C_{EP}.$$

Therefore the net catalytic flux is

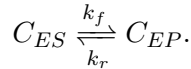
$$v_{\text{cat}} = k_f C_{ES} - k_r C_{EP}.$$

3. Binding and catalysis have different jobs

- **Binding:** redistributes E , S , and P between free and complexed states; it does not change substrate-form or product-form material.



- **Catalysis:** interconverts S -form and P -form material; it does not change the total enzyme inventory.

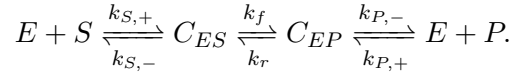


Putting the enzyme model together

The resolved enzyme has five molecular states:

$$E, \quad S, \quad P, \quad C_{ES}, \quad C_{EP}.$$

Binding allocates the enzyme among E , C_{ES} , and C_{EP} ; catalytic conversion connects the two bound states. **The complete reaction scheme is**



Three constituent totals

The enzyme, substrate-form, and product-form inventories are

$$q_E = E + C_{ES} + C_{EP}, \quad q_S = S + C_{ES}, \quad q_P = P + C_{EP}.$$

The enzyme pool q_E is constant in this model. Binding changes the individual state concentrations but does not change any of these inventories.

Catalytic net flux

$$v_{\text{cat}} = k_f C_{ES} - k_r C_{EP}.$$

Key point. Binding determines *which states are occupied*; catalysis determines *what those states do*.